

Homework 5

*Instructor: Henry Lam**Due Date: Oct 27, 2025***Problem 1** For each of the following integrals:

(a)

$$\int_0^1 \exp\{e^x\} dx$$

(b)

$$\int_{-2}^2 e^{x+x^2} dx$$

(c)

$$\int_0^\infty x(1+x^2)^{-2} dx$$

(d)

$$\int_0^\infty \int_0^{x^2} e^{-(x+y)} \sin(xy) dy dx$$

do the following: Write a pseudo-code to generate a simulation run that is appropriate for numerically approximating the integral. Then implement 1000 simulation runs on a computer. Give an estimate for the value of the integral.